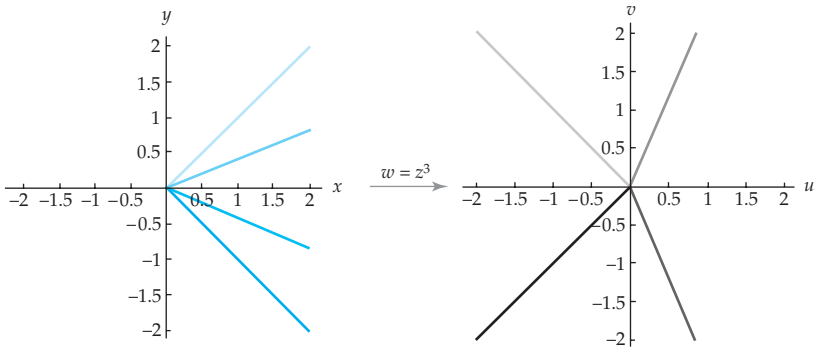
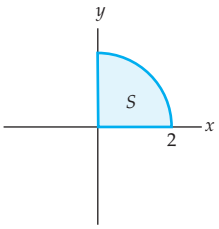
Consequently, if z and $w = z^n$ are plotted in the same copy of the complex plane, then this mapping can be visualized as the process of magnifying or contracting the modulus r of z to the modulus r^n of w, and by rotating z about the origin to increase an argument θ of z to an argument $n\theta$ of w.

We can use this description of $w = z^n$ to show that a ray emanating from the origin and making an angle of ϕ radians with the positive x-axis is mapped onto a ray emanating from the origin and making an angle of $n\phi$ radians with the positive u-axis. This property is illustrated for the mapping $w = z^3$ in FIGURE 2.4.7. Each ray shown in color in Figure 2.4.7(a) is mapped onto a ray shown in black in Figure 2.4.7(b). Since the mapping $w = z^3$ increases the argument of a point by a factor of 3, the ray nearest the x-axis in the first quadrant in Figure 2.4.7(a) is mapped onto the ray in the first quadrant in Figure 2.4.7(b), and the remaining ray in the first quadrant in Figure 2.4.7(a) is mapped onto the ray in the second quadrant in Figure 2.4.7(b). Similarly, the ray nearest the x-axis in the fourth quadrant in Figure 2.4.7(a) is mapped onto the ray in the fourth quadrant in Figure 2.4.7(b), and the remaining ray in the fourth quadrant of Figure 2.4.7(a) is mapped onto the ray in the third quadrant in Figure 2.4.7(b).

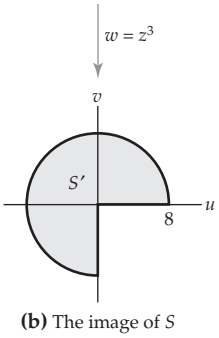


(a) Rays in the z-plane **(b)** Images of the rays in (a)

FIGURE 2.4.7 The mapping $w = z^3$



(a) The set S for Example 4



(b) The image of S

FIGURE 2.4.8 The mapping $w = z^3$

EXAMPLE 4 Image of a Circular Wedge under $w = z^3$

Determine the image of the quarter disk defined by the inequalities $|z| \leq 2$, $0 \leq \arg(z) \leq \pi/2$, under the mapping $w = z^3$.

Solution Let S denote the quarter disk and let S' denote its image under $w = z^3$. Since the moduli of the points in S vary from 0 to 2 and since the mapping $w = z^3$ cubes the modulus of a point, it follows that the moduli of the points in S' vary from $0^3 = 0$ to $2^3 = 8$. In addition, because the arguments of the points in S vary from 0 to $\pi/2$ and because the mapping $w = z^3$ triples the argument of a point, we also have that the arguments of the points in S' vary from 0 to $3\pi/2$. Therefore, S' is the set given by the inequalities $|w| \leq 8, 0 \leq \arg(w) \leq 3\pi/2$, shown in gray in FIGURE 2.4.8(b). In summary, the set S shown in color in Figure 2.4.8(a) is mapped onto the set S' shown in gray in Figure 2.4.8(b) by $w = z^3$. ☰

2.4.2 The Power Function $z^{1/n}$

We now investigate complex power functions of the form $z^{1/n}$ where n is an integer and $n \geq 2$. We begin with the case $n = 2$.

Principal Square Root Function $z^{1/2}$ In (4) of Section 1.4 we saw that the n nth roots of a nonzero complex number $z = r(\cos\theta + i\sin\theta) = re^{i\theta}$ are given by:

$$\sqrt[n]{r}\left[\cos\left(\frac{\theta + 2k\pi}{n}\right) + i\sin\left(\frac{\theta + 2k\pi}{n}\right)\right] = \sqrt[n]{r}e^{i(\theta+2k\pi)/n}$$

where $k = 0,\ 1,\ 2,\ \ldots,\ n-1$. In particular, for $n = 2$, we have that the two square roots of z are:

$$\sqrt{r}\left[\cos\left(\frac{\theta + 2k\pi}{2}\right) + i\sin\left(\frac{\theta + 2k\pi}{2}\right)\right] = \sqrt{r}e^{i(\theta+2k\pi)/2} \tag{6}$$

for $k = 0, 1$. The formula in (6) does not define a function because it assigns *two* complex numbers (one for $k = 0$ and one for $k = 1$) to the complex number z. However, by setting $\theta = \text{Arg}(z)$ and $k = 0$ in (6) we can define a function that assigns to z the unique principal square root. Naturally, this function is called the **principal square root function**.

DEFINITION 2.4.1 Principal Square Root Function

The function $z^{1/2}$ defined by:

$$z^{1/2} = \sqrt{|z|}e^{i\,\text{Arg}(z)/2} \tag{7}$$

is called the **principal square root function**.

If we set $\theta = \text{Arg}(z)$ and replace z with $re^{i\theta}$ in (7), then we obtain an alternative description of the principal square root function for $|z| > 0$:

$$z^{1/2} = \sqrt{r}e^{i\theta/2}, \quad \theta = \text{Arg}(z). \tag{8}$$

Note ▶ You should take note that the symbol $z^{1/2}$ used in Definition 2.4.1 represents something different from the same symbol used in Section 1.4. In (7) we use $z^{1/2}$ to represent the *value* of the principal square root of the complex number z, whereas in Section 1.4 the symbol $z^{1/2}$ was used to represent the *set* of two square roots of the complex number z. This repetition of notation is unfortunate, but widely used. For the most part, the context in which you see the symbol $z^{1/2}$ should make it clear whether we are referring to the principal square root or the set of square roots. In order to avoid confusion we will, at times, also explicitly state "the principal square root function $z^{1/2}$" or "the function $f(z) = z^{1/2}$ given by (7)."

EXAMPLE 5 Values of $z^{1/2}$

Find the values of the principal square root function $z^{1/2}$ at the following points:

(a) $z = 4$ **(b)** $z = -2i$ **(c)** $z = -1 + i$

Solution In each part we use (7) to determine the value of $z^{1/2}$.

(a) For $z = 4$, we have $|z| = |4| = 4$ and $\text{Arg}(z) = \text{Arg}(4) = 0$, and so from (7) we obtain:

$$4^{1/2} = \sqrt{4}e^{i(0/2)} = 2e^{i(0)} = 2.$$

(b) For $z = -2i$, we have $|z| = |-2i| = 2$ and $\text{Arg}(z) = \text{Arg}(-2i) = -\pi/2$, and so from (7) we obtain:

$$(-2i)^{1/2} = \sqrt{2}e^{i(-\pi/2)/2} = \sqrt{2}e^{-i\pi/4} = 1 - i.$$

(c) For $z = -1+i$, we have $|z| = |-1+i| = \sqrt{2}$ and $\text{Arg}(z) = \text{Arg}(-1+i) = 3\pi/4$, and so from (7) we obtain:

$$(-1+i)^{1/2} = \sqrt{(\sqrt{2})}e^{i(3\pi/4)/2} = \sqrt[4]{2}e^{i(3\pi/8)} \approx 0.4551 + 1.0987i. \quad \equiv$$

It is important that we use the *principal argument* when we evaluate the principal square root function of Definition 2.4.1. Using a different choice for the argument of z can give a different function. For example, in Section 1.4 we saw that the two square roots of i are $\frac{1}{2}\sqrt{2} + \frac{1}{2}\sqrt{2}\,i$ and $-\frac{1}{2}\sqrt{2} - \frac{1}{2}\sqrt{2}\,i$. For $z = i$ we have that $|z| = 1$ and $\text{Arg}(z) = \pi/2$. It follows from (7) that:

$$i^{1/2} = \sqrt{1}e^{i(\pi/2)/2} = 1 \cdot e^{i\,\pi/4} = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i.$$

Therefore, only the first of these two roots of i is the value of the principal square root function. Of course, we can define other "square root" functions. For example, suppose we let θ be the unique value of the argument of z in the interval $\pi < \theta \leq 3\pi$. Then $f(z) = \sqrt{|z|}e^{i\theta/2}$ defines a function for which $f(i) = -\frac{1}{2}\sqrt{2} - \frac{1}{2}\sqrt{2}i$. The function f is not the principal square root function but it is closely related. Since $\pi < \text{Arg}(z) + 2\pi \leq 3\pi$, it follows that $\theta = \text{Arg}(z) + 2\pi$, and so

$$f(z) = \sqrt{|z|}e^{i\theta/2} = \sqrt{|z|}e^{i(\text{Arg}(z)+2\pi)/2} = \sqrt{|z|}e^{i\,\text{Arg}(z)/2}e^{i\pi}.$$

Because $e^{i\pi} = -1$, the foregoing expression simplifies to $f(z) = -\sqrt{|z|}e^{i\,\text{Arg}(z)/2}$. That is, we have shown that the function $f(z) = \sqrt{|z|}e^{i\theta/2}$, $\pi < \theta \leq 3\pi$, is the negative of the principal square root function $z^{1/2}$.

A review of inverse functions can be found in Section 2.4 of the student study guide. ▶

Inverse Functions The principal square root function $z^{1/2}$ given by (7) is *an* inverse function of the squaring function z^2 examined in the first part of this section. Before elaborating on this statement, we need to review some general terminology regarding inverse functions.

A real function must be one-to-one in order to have an inverse function. The same is true for a complex function. The definition of a one-to-one complex function is analogous to that for a real function. Namely, a complex function f is **one-to-one** if each point w in the range of f is the image of a unique point z, called the *pre-image* of w, in the domain of f. That is, f is one-to-one if whenever $f(z_1) = f(z_2)$, then $z_1 = z_2$. Put another way, if $z_1 \neq z_2$, then $f(z_1) \neq f(z_2)$. This says that a one-to-one complex function will not map distinct points in the z-plane onto the same point in the w-plane. For example, the function $f(z) = z^2$ is not one-to-one because $f(i) = f(-i) = -1$. If f is a one-to-one complex function, then for any point w in the range of f there is a unique pre-image in the z-plane, which we denote by $f^{-1}(w)$. This correspondence between a point w and its pre-image $f^{-1}(w)$ defines the inverse function of a one-to-one complex function.

DEFINITION 2.4.2 Inverse Function

If f is a one-to-one complex function with domain A and range B, then the **inverse function of f**, denoted by f^{-1}, is the function with domain B and range A defined by $f^{-1}(z) = w$ if $f(w) = z$.

It follows immediately from Definition 2.4.2 that if a set S is mapped onto a set S' by a one-to-one function f, then f^{-1} maps S' onto S. In other words, the complex mappings f and f^{-1} "undo" each other. It also follows from Definition 2.4.2 that if

f has an inverse function, then $f\left(f^{-1}(z)\right) = z$ and $f^{-1}(f(z)) = z$. That is, the two compositions $f \circ f^{-1}$ and $f^{-1} \circ f$ are the identity function.

EXAMPLE 6 Inverse Function of $f(z) = z + 3i$

Show that the complex function $f(z) = z + 3i$ is one-to-one on the entire complex plane and find a formula for its inverse function.

Solution One way of showing that f is one-to-one is to show that the equality $f(z_1) = f(z_2)$ implies the equality $z_1 = z_2$. For the function $f(z) = z + 3i$, this follows immediately since $z_1 + 3i = z_2 + 3i$ implies that $z_1 = z_2$. As with real functions, the inverse function of f can often be found algebraically by solving the equation $z = f(w)$ for the symbol w. Solving $z = w + 3i$ for w, we obtain $w = z - 3i$ and so $f^{-1}(z) = z - 3i$. ☰

Solve the equation $z = f(w)$ for w to find a formula for $w = f^{-1}(z)$. ▶

From Section 2.3 we know that the mapping $f(z) = z + 3i$ in Example 6 is a translation by $3i$ and that $f^{-1}(z) = z - 3i$ is a translation by $-3i$. Suppose we represent these mappings in a single copy of the complex plane. Since translating a point by $3i$ and then translating the image by $-3i$ moves the original point back onto itself, we see that the mappings f and f^{-1} "undo" each other as expected.

Inverse Functions of z^n, $n > 2$ We now describe how to obtain an inverse function of the power function z^n, $n \geq 2$. This requires some explanation since the function $f(z) = z^n$, $n \geq 2$, is *not* one-to-one. In order to see that this is so, consider the points $z_1 = re^{i\theta}$ and $z_2 = re^{i(\theta+2\pi/n)}$ with $r \neq 0$. Because $n \geq 2$, the points z_1 and z_2 are distinct. That is, $z_1 \neq z_2$. From (5) we have that $f(z_1) = r^n e^{in\theta}$ and $f(z_2) = r^n e^{i(n\theta+2\pi)} = r^n e^{in\theta} e^{i2\pi} = r^n e^{in\theta}$. Therefore, f is not one-to-one since $f(z_1) = f(z_2)$ but $z_1 \neq z_2$. In fact, the n distinct points $z_1 = re^{i\theta}$, $z_2 = re^{i(\theta+2\pi/n)}$, $z_3 = re^{i(\theta+4\pi/n)}$, $\ldots$, $z_n = re^{i(\theta+2(n-1)\pi/n)}$ are all mapped onto the single point $w = r^n e^{in\theta}$ by $f(z) = z^n$. This fact is illustrated for $n = 6$ in FIGURE 2.4.9. The six points $z_1, z_2, \ldots, z_6$ with equal modulus and arguments that differ by $2\pi/6 = \pi/3$ shown in Figure 2.4.9 are all mapped onto the same point by $f(z) = z^6$.

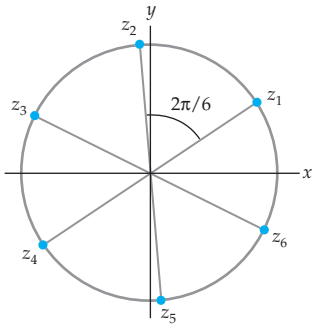
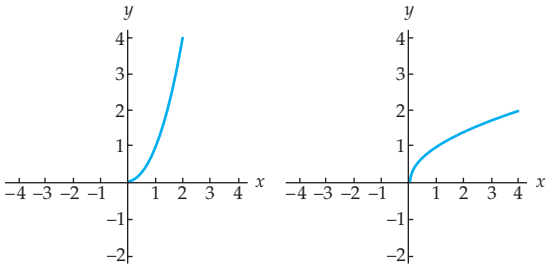


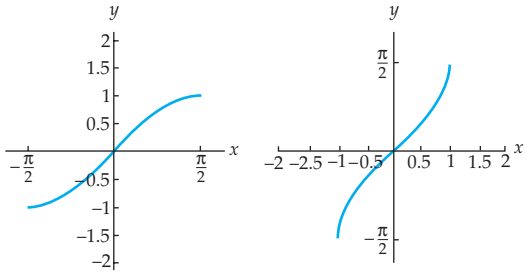
FIGURE 2.4.9 Points mapped onto the same point by $f(z) = z^6$

The preceding discussion appears to imply that the function $f(z) = z^n$, $n \geq 2$, does not have an inverse function because it is not one-to-one. You have encountered this problem before, when defining inverse functions for certain real functions in elementary calculus. For example, the real functions $f(x) = x^2$ and $g(x) = \sin x$ are not one-to-one, yet we still have the inverse functions $f^{-1}(x) = \sqrt{x}$ and $g^{-1}(x) = \arcsin x$. The key to defining these inverse functions is to appropriately restrict the domains of the functions to sets on which the functions are one-to-one. For example, whereas the real function $f(x) = x^2$ defined on the interval $(-\infty, \infty)$ is not one-to-one, the same function defined on the interval $[0, \infty)$ is one-to-one. Similarly, $g(x) = \sin x$ is not one-to-one on the interval $(-\infty, \infty)$, but it is one-to-one on the interval $[-\pi/2,\ \pi/2]$. The function $f^{-1}(x) = \sqrt{x}$ is the inverse function of the function $f(x) = x^2$ defined on the interval $[0,\ \infty)$. Since $\text{Dom}(f) = [0,\ \infty)$ and $\text{Range}(f) = [0,\ \infty)$, the domain and range of $f^{-1}(x) = \sqrt{x}$ are both $[0,\ \infty)$ as well. See FIGURE 2.4.10. In a similar manner, $g^{-1}(x) = \arcsin x$ is the inverse function of the function $g(x) = \sin x$ defined on $[-\pi/2,\ \pi/2]$. The domain and range of g^{-1} are $[-1, 1]$ and $[-\pi/2,\ \pi/2]$, respectively. See FIGURE 2.4.11. We use this same idea for the complex power function z^n, $n \geq 2$. That is, in order to define an inverse function for $f(z) = z^n$, $n \geq 2$, we must restrict the domain of f to a set on which f is a one-to-one function. One such choice of this "restricted domain" when $n = 2$ is found in Example 7.



(a) $f(x) = x^2$ **(b)** $f^{-1}(x) = \sqrt{x}$

FIGURE 2.4.10 $f(x) = x^2$ defined on $[0, \infty)$ and its inverse function



(a) $g(x) = \sin x$ **(b)** $g^{-1}(x) = \arcsin x$

FIGURE 2.4.11 $g(x) = \sin x$ defined on $[-\pi/2, \pi/2]$ and its inverse function

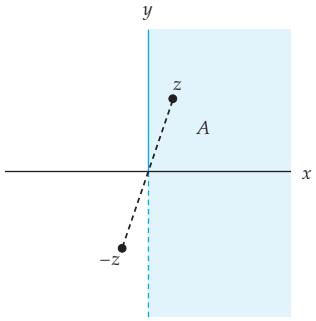


FIGURE 2.4.12 A domain on which $f(z) = z^2$ is one-to-one

EXAMPLE 7 A Restricted Domain for $f(z) = z^2$

Show that $f(z) = z^2$ is a one-to-one function on the set A defined by $-\pi/2 < \arg(z) \leq \pi/2$ and shown in color in FIGURE 2.4.12.

Solution As in Example 6, we show that f is one-to-one by demonstrating that if z_1 and z_2 are in A and if $f(z_1) = f(z_2)$, then $z_1 = z_2$. Suppose that z_1 and z_2 are in A, then we may write $z_1 = r_1 e^{i\theta_1}$ and $z_2 = r_2 e^{i\theta_2}$ with $-\pi/2 < \theta_1 \leq \pi/2$ and $-\pi/2 < \theta_2 \leq \pi/2$. If $f(z_1) = f(z_2)$, then it follows from (1) that:

$$r_1^2 e^{i2\theta_1} = r_2^2 e^{i2\theta_2}. \tag{9}$$

From (9) we conclude that the complex numbers $r_1^2 e^{i2\theta_1}$ and $r_2^2 e^{i2\theta_2}$ have the same modulus and principal argument:

$$r_1^2 = r_2^2 \quad \text{and} \quad \operatorname{Arg}\left(r_1^2 e^{i2\theta_1}\right) = \operatorname{Arg}\left(r_2^2 e^{i2\theta_2}\right). \tag{10}$$

Remember: Arg(z) is in the interval $(-\pi, \pi]$. ▶

Because both r_1 and r_2 are positive, the first equation in (10) implies that $r_1 = r_2$. Moreover, since $-\pi/2 < \theta_1 \leq \pi/2$ and $-\pi/2 < \theta_2 \leq \pi/2$, it follows that $-\pi < 2\theta_1 \leq \pi$, and $-\pi < 2\theta_2 \leq \pi$. This means that $\operatorname{Arg}\left(r_1^2 e^{i2\theta_1}\right) = 2\theta_1$ and $\operatorname{Arg}\left(r_2^2 e^{i2\theta_2}\right) = 2\theta_2$. This fact combined with the second equation in (10) implies that $2\theta_1 = 2\theta_2$, or $\theta_1 = \theta_2$. Therefore, we conclude that $z_1 = z_2$, and this proves that f is a one-to-one function on A. ☰

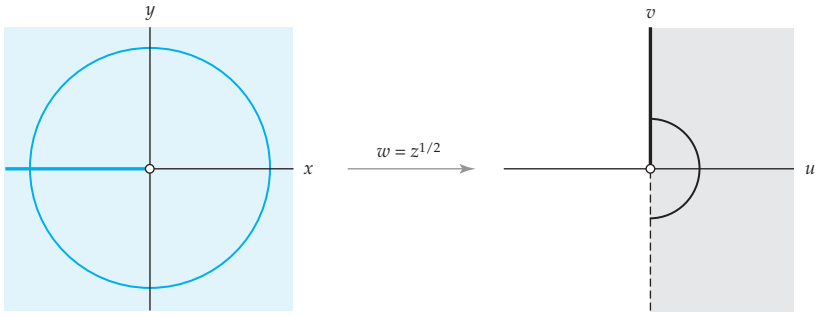
An Inverse of $f(z) = z^2$ In Example 7 we saw that the squaring function z^2 is one-to-one on the set A defined by $-\pi/2 < \arg(z) \leq \pi/2$. It follows from Definition 2.4.2 that this function has a well-defined inverse function f^{-1}. We now proceed to show that this inverse function is the principal square root function $z^{1/2}$ from Definition 2.4.1. In order to do so, let $z = re^{i\theta}$ and $w = \rho e^{i\phi}$ where θ and ϕ are the principal arguments of z and w respectively. Suppose that $w = f^{-1}(z)$. Since

the range of f^{-1} is the domain of f, the principal argument ϕ of w must satisfy:

$$-\frac{\pi}{2} < \phi \leq \frac{\pi}{2}. \tag{11}$$

On the other hand, $f(w) = w^2 = z$, and so w is one of the two square roots of z given by (6). That is, either $w = \sqrt{r}e^{i\theta/2}$ or $w = \sqrt{r}e^{i(\theta+2\pi)/2}$. Because $-\pi < \theta \leq \pi$, we have $-\pi/2 < \theta/2 \leq \pi/2$ and $\pi/2 < (\theta + 2\pi)/2 \leq 3\pi/2$. This means that $\text{Arg}(\sqrt{r}e^{i\theta/2})$ is in the interval $(-\pi/2, \pi/2]$ whereas $\text{Arg}(\sqrt{r}e^{i(\theta+2\pi)/2})$ is either in the interval $(\pi/2, \pi]$ or $(-\pi, -\pi/2]$. Therefore, in order for $\text{Arg}(w) = \phi$ to satisfy (11), we must have $w = \sqrt{r}e^{i\theta/2}$, which is the value of the principal square root function $z^{1/2}$ given by (8).

From Definition 2.4.1, the principal square root function is only defined for $z \neq 0$ (remember that Arg(0) is undefined). That is, the domain of $z^{1/2}$ is the complex plane **C** excluding 0. Since $z^{1/2}$ and z^2 defined on the set $-\pi/2 < \arg(z) \leq \pi/2$ are inverse functions, it follows that the range of $z^{1/2}$ is just the (restricted) domain of z^2. Therefore, the range of $z^{1/2}$ is the set A defined by $-\pi/2 < \arg(w) \leq \pi/2$. These observations regarding the domain and range of the principal square root function are depicted in Figure 2.4.3. The mapping $w = z^{1/2}$ takes the domain shown in color in **FIGURE 2.4.13(a)** onto the set shown in gray in Figure 2.4.13(b). It is important to compare the mappings depicted in Figures 2.4.3 and 2.4.13. Whereas the function z^2 *in general* maps both of the rays $\arg(z) = \pi/2$ and $\arg(z) = -\pi/2$ onto the negative real axis, the function $z^{1/2}$ maps the negative real axis only to the ray $\arg(w) = \pi/2$. This is because the ray $\arg(z) = \pi/2$ is the unique choice of these two rays that lies in the restricted domain for z^2 used to define its inverse function $z^{1/2}$.



(a) Domain of $z^{1/2}$ **(b)** Range of $z^{1/2}$

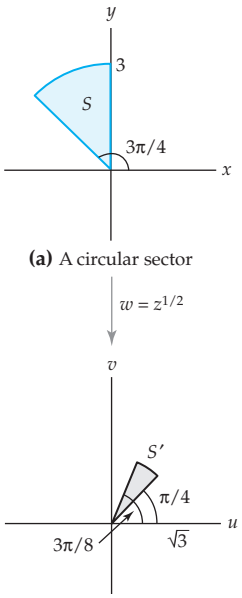
FIGURE 2.4.13 The principal square root function $w = z^{1/2}$

The Mapping $w = z^{1/2}$ As a mapping, the function z^2 squares the modulus of a point and doubles its argument. Because the principal square root function $z^{1/2}$ is an inverse function of z^2, it follows that the mapping $w = z^{1/2}$ takes the square root of the modulus of a point and halves its principal argument. That is, if $w = z^{1/2}$, then we have $|w| = \sqrt{|z|}$ and $\text{Arg}(w) = \frac{1}{2}\text{Arg}(z)$. These relations follow directly from (7) and are helpful in determining the images of sets under $w = z^{1/2}$.

EXAMPLE 8 Image of a Circular Sector under $w = z^{1/2}$

Find the image of the set S defined by $|z| \leq 3$, $\pi/2 \leq \arg(z) \leq 3\pi/4$, under the principal square root function.

Solution Let S' denote the image of S under $w = z^{1/2}$. Since $|z| \leq 3$ for points in S and since $z^{1/2}$ takes the square root of the modulus of a point, we must have that



(a) A circular sector

(b) The image of the set in (a)

FIGURE 2.4.14 The mapping $w = z^{1/2}$

$|w| \leq \sqrt{3}$ for points w in S'. In addition, since $\pi/2 \leq \arg(z) \leq 3\pi/4$ for points in S and since $z^{1/2}$ halves the argument of a point, it follows that $\pi/4 \leq \arg(w) \leq 3\pi/8$ for points w in S'. Therefore, we have shown that the set S shown in color in **FIGURE 2.4.14(a)** is mapped onto the set S' shown in gray in Figure 2.4.14(b) by $w = z^{1/2}$. ☰

Principal *n*th Root Function By modifying the argument given in Example 7 that the function $f(z) = z^2$ is one-to-one on the set defined by $-\pi/2 < \arg(z) \leq \pi/2$, we can show that the complex power function $f(z) = z^n$, $n > 2$, is one-to-one on the set defined by

$$-\frac{\pi}{n} < \arg(z) \leq \frac{\pi}{n}. \tag{12}$$

It is also relatively easy to see that the image of the set defined by (12) under the mapping $w = z^n$ is the entire complex plane **C** excluding $w = 0$. Therefore, there is a well-defined inverse function for f. Analogous to the case $n = 2$, this inverse function of z^n is called the **principal *n*th root function** $z^{1/n}$. The domain of $z^{1/n}$ is the set of all nonzero complex numbers, and the range of $z^{1/n}$ is the set of complex numbers w satisfying $-\pi/n < \arg(w) \leq \pi/n$. A purely algebraic description of the principal nth root function is given by the following formula, which is analogous to (7).

DEFINITION 2.4.3 Principal *n*th Root Functions

For $n \geq 2$, the function $z^{1/n}$ defined by:

$$z^{1/n} = \sqrt[n]{|z|}\, e^{i\,\mathrm{Arg}(z)/n} \tag{13}$$

is called the **principal *n*th root function**.

Notice that the principal square root function $z^{1/2}$ from Definition 2.4.1 is simply a special case of (13) with $n = 2$. Notice also that in Definition 2.4.3 we use the symbol $z^{1/n}$ to represent something different than the same symbol used in Section 1.4. As with the symbol $z^{1/2}$, whether $z^{1/n}$ represents the principal nth root or the set of principal nth roots will be clear from the context or stated explicitly.

By setting $z = re^{i\theta}$ with $\theta = \mathrm{Arg}(z)$ we can also express the principal nth root function as:

$$z^{1/n} = \sqrt[n]{r}\, e^{i\theta/n}, \quad \theta = \mathrm{Arg}(z). \tag{14}$$

EXAMPLE 9 Values of $z^{1/n}$

Find the value of the given principal nth root function $z^{1/n}$ at the given point z.

(a) $z^{1/3}$; $z = i$ **(b)** $z^{1/5}$; $z = 1 - \sqrt{3}i$

Solution In each part we use (13).

(a) For $z = i$, we have $|z| = 1$ and $\mathrm{Arg}(z) = \pi/2$. Substituting these values in (13) with $n = 3$ we obtain:

$$i^{1/3} = \sqrt[3]{1}\, e^{i(\pi/2)/3} = e^{i\pi/6} = \frac{\sqrt{3}}{2} + \frac{1}{2}i.$$

(b) For $z = 1 - \sqrt{3}i$, we have $|z| = 2$ and $\mathrm{Arg}(z) = -\pi/3$. Substituting these values in (13) with $n = 5$ we obtain:

$$\left(1 - \sqrt{3}i\right)^{1/5} = \sqrt[5]{2}\, e^{i(-\pi/3)/5} = \sqrt[5]{2}\, e^{-i(\pi/15)} \approx 1.1236 - 0.2388i. \quad ☰$$

Multiple-Valued Functions In Section 1.4 we saw that a nonzero complex number z has n distinct nth roots in the complex plane. This means that the process of "taking the nth root" of a complex number z *does not* define a complex function because it assigns a set of n complex numbers to the complex number z. We introduced the symbol $z^{1/n}$ in Section 1.4 to represent the set consisting of the n nth roots of z. A similar type of process is that of finding the argument of a complex number z. Because the symbol $\arg(z)$ represents an infinite set of values, it also does not represent a complex function. These types of operations on complex numbers are examples of **multiple-valued functions**. This term often leads to confusion since a multiple-valued function is *not* a function; a function, by definition, must be *single-valued*. However unfortunate, the term *multiple-valued function* is a standard one in complex analysis and so we shall use it from this point on. We will adopt the following functional notation for multiple-valued functions.

Notation: Multiple-Valued Functions

When representing multiple-valued functions with functional notation, we will use uppercase letters such as $F(z) = z^{1/2}$ or $G(z) = \arg(z)$. Lower-case letters such as f and g will be reserved to represent functions.

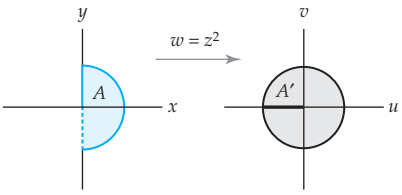
This notation will help avoid confusion associated to the symbols like $z^{1/n}$. For example, we should assume that $g(z) = z^{1/3}$ refers to the principal cube root function defined by (13) with $n = 3$, whereas, $G(z) = z^{1/3}$ represents the multiple-valued function that assigns the three cube roots of z to the value of z. Thus, we have $g(i) = \frac{1}{2}\sqrt{3} + \frac{1}{2}i$ from Example 9(a) and $G(i) = \left\{\frac{1}{2}\sqrt{3} + \frac{1}{2}i, -\frac{1}{2}\sqrt{3} + \frac{1}{2}i, -i\right\}$ from Example 1 of Section 1.4.

Remarks

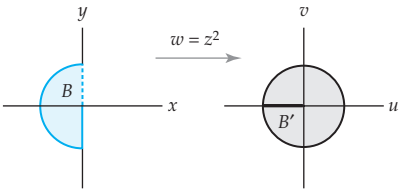
(*i*) In (5) in Exercises 1.4 we defined a rational power of z. One way to define a power function $z^{m/n}$ where m/n is a rational number is as a composition of the principal nth root function and the power function z^m. That is, we can define $z^{m/n} = \left(z^{1/n}\right)^m$. Thus, for $m/n = 2/3$ and $z = 8i$ you should verify that $(8i)^{2/3} = 2 + 2\sqrt{3}i$. Of course, using a root other than the principal nth root gives a possibly different function.

(*ii*) You have undoubtedly noticed that the complex linear mappings studied in Section 2.3 are much easier to visualize than the mappings by complex power functions studied in this section. In part, mappings by complex power functions are more intricate because these functions are not one-to-one. The visualization of a complex mapping that is **multiple-to-one** is difficult and it follows that the multiple-valued functions, which are "inverses" to the multiple-to-one functions, are also hard to visualize. A technique attributed to the mathematician Bernhard Riemann (1826–1866) for visualizing multiple-to-one and multiple-valued functions is to construct a **Riemann surface** for the mapping. Since a rigorous description of Riemann surfaces is beyond the scope of this text, our discussion of these surfaces will be informal.

We begin with a description of a Riemann surface for the complex squaring function $f(z) = z^2$ defined on the closed unit disk $|z| \leq 1$.



(a) A maps onto A'

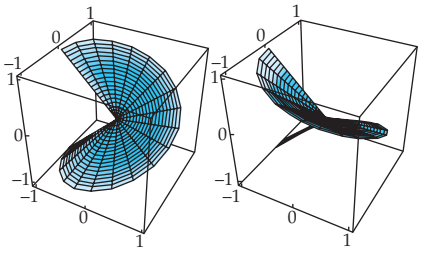


(b) B maps onto B'

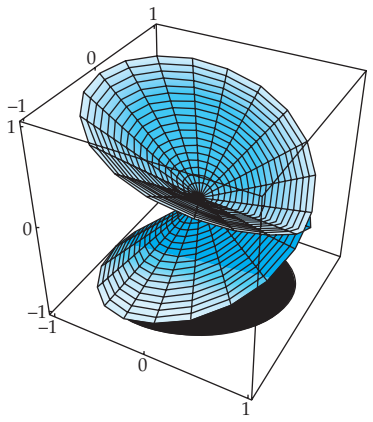
FIGURE 2.4.15 The mapping $w = z^2$



FIGURE 2.4.16 The cut disks A' and B'



(a) The cut disks A' and B' in xyz-space

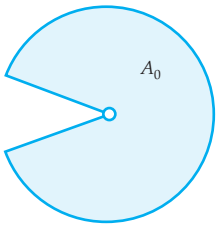


(b) Riemann surface in xyz-space

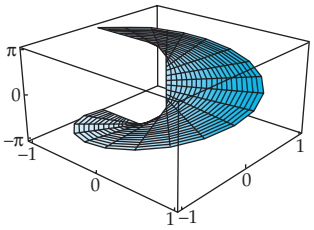
FIGURE 2.4.17 A Riemann surface for $f(z) = z^2$

On page 80 we saw follows from Example 7 that $f(z) = z^2$ is one-to-one on the set A defined by $|z| \leq 1$, $-\pi/2 < \arg(z) \leq \pi/2$. Under the complex mapping $w = z^2$, the set A shown in color in **FIGURE 2.4.15(a)** is mapped onto the closed unit disk $|w| \leq 1$ shown in gray in Figure 2.4.15(a). In a similar manner, we can show that $w = z^2$ is a one-to-one mapping of the set B defined by $|z| \leq 1$, $\pi/2 < \arg(z) \leq 3\pi/2$, onto the closed unit disk $|w| \leq 1$. See Figure 2.4.15(b). Since the unit disk $|z| \leq 1$ is the union of the sets A and B, the image of the disk $|z| \leq 1$ under $w = z^2$ covers the disk $|w| \leq 1$ twice (once by A and once by B). We visualize this "covering" by considering *two* image disks for $w = z^2$. Let A' denote the image of A under f and let B' denote the image of B under f. See Figure 2.4.15. Now imagine that the disks A' and B' have been cut open along the negative real axis as shown in **FIGURE 2.4.16**. The edges shown in black in Figure 2.4.16 of the cut disks A' and B' are the images of edges shown in color of A and B, respectively. Similarly, the dashed edges of A' and B' shown in Figure 2.4.16 are the images of the dashed edges of A and B, respectively. We construct a Riemannn surface for $f(z) = z^2$ by stacking the cut disks A' and B' one atop the other in xyz-space and attaching them by gluing together their edges. The black edge of A' shown in Figure 2.4.16 is glued to the dashed edge of B' shown in Figure 2.4.16, and the dashed edge of A' is glued to the black edge of B'. After attaching in this manner we obtain the Riemann surface shown in **FIGURE 2.4.17**. We assume that this surface is lying directly above the closed unit disk $|w| \leq 1$. Although $w = z^2$ is not a one-to-one mapping of the closed unit disk $|z| \leq 1$ onto the closed unit disk $|w| \leq 1$, it is a one-to-one mapping of the closed unit disk $|z| \leq 1$ onto the Riemann surface that we have constructed. Moreover, the two-to-one covering nature of the mapping $w = z^2$ can be visualized by mapping the disk $|z| \leq 1$ onto the Riemann surface, then projecting the points of the Riemann surface vertically down onto the disk $|w| \leq 1$. In addition, by reversing the order in this procedure, we can also use this Riemann surface to help visualize the multiple-valued function $F(z) = z^{1/2}$.

Another interesting Riemann surface is one for the multiple-valued function $G(z) = \arg(z)$ defined on the punctured disk $0 < |z| \leq 1$. To construct this surface, we take a copy of the punctured disk $0 < |z| \leq 1$ and cut it open along the negative real axis. See **FIGURE 2.4.18(a)**. Call this cut disk A_0 and let it represent the points in the domain written in exponential notation as $re^{i\theta}$ with $-\pi < \theta \leq \pi$. Take another copy of the cut disk, call it A_1, and let it represent the points in the domain written as $re^{i\theta}$ with $\pi < \theta \leq 3\pi$. Similarly, let A_{-1} be a cut disk that represents the points in the domain written as $re^{i\theta}$ with $-3\pi < \theta \leq -\pi$. Continue in this manner to produce an infinite set of cut disks $\ldots$ A_{-2}, A_{-1}, A_0, A_1, A_2 $\ldots$. In general, the cut disk A_n represents points in the domain of G expressed as $re^{i\theta}$ with $(2n-1)\pi < \theta \leq (2n+1)\pi$. Now place each disk A_n in xyz-space so that the point $re^{i\theta}$ with $(2n-1)\pi < \theta \leq (2n+1)\pi$ lies at height θ directly above the point $re^{i\theta}$ in the xy-plane. The cut disk A_0 placed in xyz-space is shown in Figure 2.4.18(b). The collection of all the cut disks in xyz-space forms the Riemann surface for the multiple-valued function $G(z) = \arg(z)$ shown in **FIGURE 2.4.19**. The Riemann surface indicates how this multiple-valued function maps the punctured disk $0 < |z| \leq 1$ onto the real line. Namely, a vertical line passing through any point z in $0 < |z| \leq 1$ intersects this Riemann surface at infinitely many points. The heights of the points of intersection represent different choices for the argument of z. Therefore, by horizontally projecting the points of intersection onto the vertical axis, we see the infinitely many images of $G(z) = \arg(z)$.



(a) The cut disk A_0



(b) A_0 placed in xyz-space

FIGURE 2.4.18 The cut disk A_0

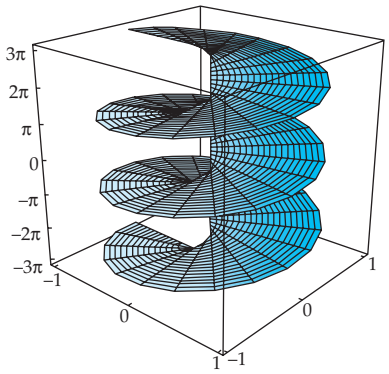


FIGURE 2.4.19 The Riemann surface for $G(z) = \arg(z)$

EXERCISES 2.4

Answers to selected odd-numbered problems begin on page ANS-8.

2.4.1 The Power Function z^n

In Problems 1–14, find the image of the given set under the mapping $w = z^2$. Represent the mapping by drawing the set and its image.

1. the ray $\arg(z) = \dfrac{\pi}{3}$

2. the ray $\arg(z) = -\dfrac{3\pi}{4}$

3. the line $x = 3$

4. the line $y = -5$

5. the line $y = -\dfrac{1}{4}$

6. the line $x = \dfrac{3}{2}$

7. the positive imaginary axis

8. the line $y = x$

9. the circular arc $|z| = \dfrac{1}{2}, 0 \le \arg(z) \le \pi$

10. the circular arc $|z| = \dfrac{4}{3}, -\dfrac{\pi}{2} \le \arg(z) \le \dfrac{\pi}{6}$

11. the triangle with vertices 0, 1, and $1 + i$

12. the triangle with vertices 0, $1 + 2i$, and $-1 + 2i$

13. the square with vertices 0, 1, $1 + i$, and i

14. the polygon with vertices 0, 1, $1 + i$, and $-1 + i$

In Problems 15–20, find the image of the given set under the given composition of a linear function with the squaring function.

15. the ray $\arg(z) = \dfrac{\pi}{3}$; $f(z) = 2z^2 + 1 - i$

16. the line segment from 0 to $-1+ i$; $f(z) = \sqrt{2}z^2 + 2 - i$

17. the line $x = 2$; $f(z) = iz^2 - 3$

18. the line $y = -3$; $f(z) = -z^2 + i$

19. the circular arc $|z| = 2, 0 \le \arg(z) \le \dfrac{\pi}{2}$; $f(z) = \frac{1}{4}e^{i\pi/4}z^2$

20. the triangle with vertices 0, 1, and $1 + i$; $f(z) = -\frac{1}{4}iz^2 + 1$

21. Find the image of the ray $\arg(z) = \pi/6$ under each of the following mappings.

(a) $f(z) = z^3$ **(b)** $f(z) = z^4$ **(c)** $f(z) = z^5$

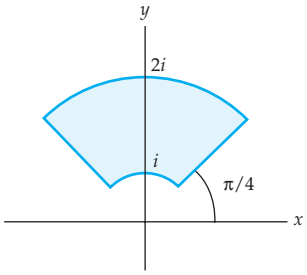


FIGURE 2.4.20 Figure for Problems 23 and 24

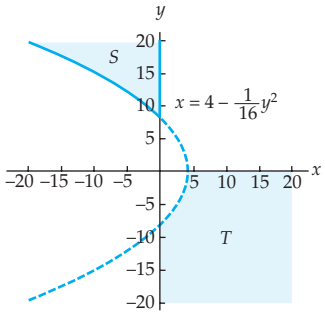


FIGURE 2.4.21 Figure for Problem 39

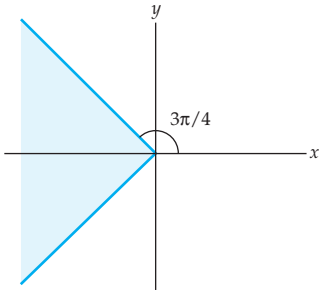


FIGURE 2.4.22 Figure for Problem 40

22. Find the image of the first quadrant of the complex plane under each of the following mappings.

(a) $f(z) = z^2$ (b) $f(z) = z^3$ (c) $f(z) = z^4$

23. Find the image of the region $1 \le |z| \le 2$, $\pi/4 \le \arg(z) \le 3\pi/4$, shown in **FIGURE 2.4.20** under each of the following mappings.

(a) $f(z) = z^2$ (b) $f(z) = z^3$ (c) $f(z) = z^4$

24. Find the image of the region shown in Figure 2.4.20 under each of the following mappings.

(a) $f(z) = 3z^2 + i$ (b) $f(z) = (i+1)z^3 + 1$ (c) $f(z) = \frac{1}{2}z^4 - i$

2.4.2 The Power Function $z^{1/n}$

In Problems 25–30, use (13) to find the value of the given principal nth root function at the given value of z.

25. $z^{1/2}$, $z = -i$

26. $z^{1/2}$, $z = 2 + i$

27. $z^{1/3}$, $z = -1$

28. $z^{1/3}$, $z = -3 + 3i$

29. $z^{1/4}$, $z = -1 + \sqrt{3}i$

30. $z^{1/5}$, $z = -4\sqrt{3} + 4i$

In Problems 31–38, find the image of the given set under the principal square root mapping $w = z^{1/2}$. Represent the mapping by drawing the set and its image.

31. the ray $\arg(z) = \dfrac{\pi}{4}$

32. the ray $\arg(z) = -\dfrac{2\pi}{3}$

33. the positive imaginary axis

34. the negative real axis

35. the arc $|z| = 9$, $-\dfrac{\pi}{2} \le \arg(z) \le \pi$

36. the arc $|z| = \dfrac{4}{7}$, $-\dfrac{\pi}{2} \le \arg(z) \le \dfrac{\pi}{4}$

37. the parabola $x = \dfrac{9}{4} - \dfrac{y^2}{9}$

38. the parabola $x = \dfrac{y^2}{10} - \dfrac{5}{2}$

39. (a) Find the image of the region S shown in **FIGURE 2.4.21** under the principal square root mapping $w = z^{1/2}$.

(b) Find the image of the region T shown in Figure 2.4.21 under the principal square root mapping $w = z^{1/2}$.

40. Find the image of the region shown in **FIGURE 2.4.22** under the principal square root function $w = z^{1/2}$. (Be careful near the negative real axis!)

Focus on Concepts

41. Use a procedure similar to that used in Example 2 to find the image of the hyperbola $xy = k$, $k \neq 0$, under $w = z^2$.

42. Use a procedure similar to that used in Example 2 to find the image of the hyperbola $x^2 - y^2 = k$, $k \neq 0$, under the mapping $w = z^2$.

43. Find two sets in the complex plane that are mapped onto the ray $\arg(w) = \pi/2$ by the function $w = z^2$.

44. Find two sets in the complex plane that are mapped onto the set bounded by the curves $u = -v$, $u = 1 - \frac{1}{4}v^2$, and the real axis $v = 0$ by the function $w = z^2$.

45. In Example 2 it was shown that the image of a vertical line $x = k$, $k \neq 0$, under $w = z^2$ is the parabola $u = k^2 - \dfrac{v^2}{4k^2}$. Use this result, your knowledge of linear mappings, and the fact that $w = -(iz)^2$ to prove that the image of a horizontal line $y = k$, $k \neq 0$, is the parabola $u = -\left(k^2 - \dfrac{v^2}{4k^2}\right)$.

46. Find three sets in the complex plane that map onto the set $\arg(w) = \pi$ under the mapping $w = z^3$.